

Akul Ramasubban

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Education

Purdue University

B.S. in Mechanical Engineering- Presidential Scholar

West Lafayette, IN

May 2027

Technical Experience

Tesla

Cell Manufacturing Equipment/Process Design Engineering Intern

January 2026 - August 2026

- Improved tool availability ~**10%** by building an accelerated-test jig to match vibration frequency of the tool to simulate long-term failure of fastening mechanism and reduce future shutdowns due to quality issues.
- Engineered and deployed an autonomous production tool with timed actuation to correct misaligned current collectors from an inconsistent supplier process, eliminating the floor's largest availability/IEE loss and scrap.
- Conducted RCA on cell laser welding process suffering significant yield loss; applied DOE and JMP-driven insights to implement production changes that improved downstream cell yield by ~**10,000** cells/week.
- Designed and implemented an equipment upgrade that sharply reduced cell contamination, while simultaneously reducing maintenance downtime by **25%**, helping increase cell production targets to support cybercab production.
- Created and deployed a protective barrier that prevented equipment failure on production equipment using production trials and rapid prototyping, leading to an availability improvement of **> 2.5%** and **\$1000** per failure.
- Led a fleetwide effort to standardize laser weld power and performed recalibration on each tool to reduce the occurrence of asymmetric weld strength by **20%** on average, improving equipment reliability and consistency.

Ford Motor Company

Product Development Intern- Forward Model

May 2025 - August 2025

- Facilitated a Six Sigma 3L5Y root-cause panel to resolve a faulty fuel-injector tip; implemented supplier pre-ship containment that prevented the defect from escalating to a field issue or NHTSA recall.
- Improved model-based design process using MagicDraw to accelerate DFMEA development for engine systems.
- Mapped real-world customer usage and refined test methods, increasing in-field quality by **20%**.
- Redesigned operator instruction sheet after multiple plant Gemba visits, boosting UAW operator efficiency.

Purdue Electric Racing (Formula SAE EV)

Ex-Module Lead/ Senior Battery Advisor

August 2023 - Present

- Designed a waterproof battery enclosure that achieved **100%** leak-free performance during tech inspection.
- Engineered a Simscape battery model to improve peak battery performance and increase pack efficiency.
- Led FEA-validated module design for space optimization, increasing energy density by **30%** in new architecture.
- Constructed a thermal interface and spacer system to improve cooling, enhancing battery performance by **10%**.
- Developed high-accuracy cooling plate validation test for edge cases using Arduino, contributing to top 10 finish.

Purdue Humanoid Robot Club

Quadruped Project Lead

August 2026 - Present

- Lead team building vision-based quadruped that can detect and jump on raised platforms with high-torque motors.
- Designed and fabricated prototype within 1 month, developing spring-assisted and linkage-driven leg concepts.
- Ran MuJoCo and Gazebo simulations to compare leg designs for jumping performance and landing stability.

Purdue MARS (Mechanisms and Robotic Systems) Lab

Undergraduate Researcher- NSF funded project

August 2026 - Present

- Designing robotic arm with actuated laser end effector, mounted on quadruped for targeted apple disease treating.
- Building disease detection pipeline using DINO and SAM to identify and segment infected apple leaves and fruit.
- Integrating perception with arm control to treat only infected regions, minimizing pesticide use and crop loss.

Tredence Inc.

Data Science Intern

May 2024 - August 2024

- Identified sector trends using NumPy and Matplotlib libraries in Python to inform retail decisions for clients.
- Devised multiple regression models to analyze supply chain data, reducing costs for key customers by **15%**.
- Leveraged data cleaning, analysis, and visualization skills using SQL to drive scalable insights and strategies.

Skills

C | MATLAB | DFMEA | SolidWorks | FEA | ROS2 | SQL | Python | GD&T | SPC | 3DX | DFM | Simulink | 3D printing